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Inventor(s)	Buda; Yossef Israel et al.

System and method for determining location and orientation of an object in a space

Abstract

A system and method for determining location and/or orientation of a sensor may include, stored in a database a representation of an element in a first space. A mapping between the representation and input from a first sensor may be created. Using the mapping and input from a second sensor in a second space, one or more elements in the database may be identified. A location and/or orientation of the second sensor in the second space may be determined based on the one or more elements.

Inventors:	**Buda; Yossef Israel (Kfar-Adumim, IL), Israel; Tal (Mazkeret-Batya, IL)**
Applicant:	**CEPTION TECHNOLOGIES LTD.** (Jerusalem, IL)
Family ID:	**66768888**
Assignee:	**CEPTION TECHNOLOGIES LTD. (Jerusalem, IL)**
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Primary Examiner: Tran; Khoi H

Assistant Examiner: Peche; Jorge O

Attorney, Agent or Firm: AlphaPatent Associates Ltd.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a US National Phase Application of PCT International Patent Application No. PCT/IL2020/050401 International Filing Date Apr. 2, 2020, claiming the benefit of IL Patent Application No. 265818 filed Apr. 3, 2019, which is hereby incorporated by reference.

FIELD OF THE INVENTION

(2) The present invention relates generally to determining location and orientation of an object in a space. More specifically, the present invention relates to determining location and orientation using data from a plurality of modalities.

BACKGROUND OF THE INVENTION

(3) Systems and methods for determining a position or location of an object are known in the art, e.g., Global Positioning System (GPS) and Global Navigation Satellite Systems (GNSS). However,

known systems and methods suffer from a number of drawbacks.

(4) For example, the accuracy of known systems is limited to a few meters at best. Additionally, to function properly, known systems require continuous connection between at least two systems, e.g., between a smartphone and a number of satellites. Moreover, known systems and methods do not enable determining a location based on input from a sensor or modality other than a predefined modality. For example, a GPS system cannot determine a location based on images of objects in a space.

SUMMARY OF THE INVENTION

(5) In some embodiments, a representation of an element in a first space may be created and stored in a database. A mapping between the representation and input from a first perception sensor may be created. Using the mapping, and input from a second sensor in a second space, one or more elements in the database may be identified, located or extracted and, using attributes of the one or more elements a location and/or orientation of the second perception sensor in the second space may be determined.

(6) A first mapping may be according to a first condition and a second mapping may be according to a second condition. A mapping may be selected based on a condition related to a sensor. A representation of an element in the database may be based on a first point of view and input from a sensor may be from, or related to, a second, different point of view.

(7) A plurality of mappings for a respective plurality of sensor types may be created. A mapping may be selected based on a type of a sensor. A subset of a set of elements in a database may be selected based on a type of a sensor. A representation of an element may be updated in a database based on input from a sensor. An element represented in a database may be sensed by at least one of: light signals, audio signals, heat, moisture level and electromagnetic waves.

(8) An element in a database may be created based on at least one of: an image, a construction plan, a road map, a blue print, a satellite view, a street view, a top view, a side view an architectural plan and vector data. Information included or stored in the database may be related to, and the location and orientation of a sensor in a space may be determined according to, at least one of: a local coordinate system, a common coordinate system and a global coordinate system.

(9) Information in a database for matching with input from a sensor may be selected based on received or calculated location information. An estimation of a location of a sensor in a space may be determined based on location information related to the sensor. An element may be created based on raw data received from a sensor.

(10) An embodiment may include encoding data from a plurality of sources to produce a unified format representation of the data and including the unified format representation in the database. An embodiment may include utilizing machine learning to identify a set of elements in the database and matching the input with the set of elements. An embodiment may include determining location and orientation of a three dimensional (3D) space by: obtaining a 3D representation of the space using input from a mobile sensor; and determining a location and orientation of the space by correlating the 3D representation with a top view representation of the space.

(11) An embodiment may include projecting a 3D representation on a top view representation. An embodiment may include receiving a set of images and using a relative location in at least some of the images to generate a 3D representation. A top view on which a 3D representation is projected may include at least one of: an aerial image, a satellite image, a structural map, a road map and a digital elevation model (DEM). A portion of a top view representation may be selected based on location information of a space.

(12) A relative location of elements in a space may be determined based on at least one of: input from a sensor and processing a set of inputs having a respective set of points of view. Other aspects and/or advantages of the present invention are described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting examples of embodiments of the disclosure are described below with reference to figures attached hereto that are listed following this paragraph. Identical features that appear in more than one figure are generally labeled with a same label in all the figures in which they appear. A label labeling an icon representing a given feature of an embodiment of the disclosure in a figure may be used to reference the given feature. Dimensions of features shown in the figures are chosen for convenience and clarity of presentation and are not necessarily shown to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity, or several physical components may be included in one functional block or element. Further, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements.

(2) The subject matter regarded as the invention is particularly pointed out and distinctly claimed in the concluding portion of the specification. The invention, however, both as to organization and method of operation, together with objects, features and advantages thereof, may best be understood by reference to the following detailed description when read with the accompanied drawings. Embodiments of the invention are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like reference numerals indicate corresponding, analogous or similar elements, and in which:

(3) FIG. **1** shows a block diagram of a computing device according to illustrative embodiments of the present invention;

(4) FIG. **2** is an overview of a system according to illustrative embodiments of the present invention;

(5) FIG. **3** is an overview of a system and flow according to illustrative embodiments of the present invention; and

(6) FIG. **4** shows a flowchart of a method according to illustrative embodiments of the present invention.

DETAILED DESCRIPTION

(7) In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the invention. However, it will be understood by those skilled in the art that the present invention may be practiced without these specific details. In other instances, well-known methods, procedures, and components, modules, units and/or circuits have not been described in detail so as not to obscure the invention. Some features or elements described with respect to one embodiment may be combined with features or elements described with respect to other embodiments. For the sake of clarity, discussion of same or similar features or elements may not be repeated.

(8) Although embodiments of the invention are not limited in this regard, discussions utilizing terms such as, for example, “processing,” “computing,” “calculating,” “determining,” “establishing”, “analyzing”, “checking”, or the like, may refer to operation(s) and/or process(es) of a computer, a computing platform, a computing system, or other electronic computing device, that manipulates and/or transforms data represented as physical (e.g., electronic) quantities within the computer's registers and/or memories into other data similarly represented as physical quantities within the computer's registers and/or memories or other information non-transitory storage medium that may store instructions to perform operations and/or processes. Although embodiments of the invention are not limited in this regard, the terms “plurality” and “a plurality” as used herein may include, for example, “multiple” or “two or more”. The terms “plurality” or “a plurality” may be used throughout the specification to describe two or more components, devices, elements, units, parameters, or the like. The term set when used herein may include one or more items.

(9) Unless explicitly stated, the method embodiments described herein are not constrained to a particular order in time or to a chronological sequence. Additionally, some of the described method elements can occur, or be performed, simultaneously, at the same point in time, or concurrently. Some of the described method elements may be skipped, or they may be repeated, during a sequence of operations of a method.
(10) Reference is made to FIG. **1**, showing a non-limiting, block diagram of a computing device or system **100** that may be used to determine location and/or orientation of a sensor (or a device connected to the sensor) according to some embodiments of the present invention. Computing device **100** may include a processor or controller **105** that may be a hardware controller. For example, computer hardware processor or hardware controller **105** may be, or may include, a central processing unit processor (CPU), a chip or any suitable computing or computational device. Computing system **100** may include a memory **120**, executable code **125**, a storage system **130** and input/output (I/O) components **135**. Controller or processor **105** (or one or more controllers or processors, possibly across multiple units or devices) may be configured (e.g., by executing software or code) to carry out methods described herein, and/or to execute or act as the various modules, units, etc., for example by executing software or by using dedicated circuitry. More than one computing devices **100** may be included in, and one or more computing devices **100** may be, or act as the components of, a system according to some embodiments of the invention.
(11) Memory **120** may be a hardware memory. For example, memory **120** may be, or may include machine-readable media for storing software e.g., a Random-Access Memory (RAM), a read only memory (ROM), a memory chip, a Flash memory, a volatile and/or non-volatile memory or other suitable memory units or storage units. Memory **120** may be or may include a plurality of, possibly different memory units. Memory **120** may be a computer or processor non-transitory readable medium, or a computer non-transitory memory device or storage medium, e.g., a RAM. Some embodiments may include a non-transitory storage medium having stored thereon instructions which when executed cause the processor to carry out methods disclosed herein.
(12) Executable code **125** may be an application, a program, a process, task or script. A program, application or software as referred to herein may be any type of instructions, e.g., firmware, middleware, microcode, hardware description language etc. that, when executed by one or more hardware processors or controllers **105**, cause a processing system or device (e.g., system **100**) to perform the various functions described herein.
(13) Executable code **125** may be executed by controller or processor **105** possibly under control of an operating system. For example, executable code **125** may be an application that matches features or elements sensed by a sensor to element or features in a database as further described herein. Executable code **125** may be an application that identifies or extracts features or elements from a databases and associates the features or elements with signatures, identifiers or descriptions thus enabling a mapping between inputs from a variety of sensors and the features in the database.
(14) Although, for the sake of clarity, a single item of executable code **125** is shown in FIG. **1**, a system according to some embodiments of the invention may include a plurality of executable code segments similar to executable code **125** that may be loaded into memory **120** and cause controller or processor **105** to carry out methods described herein. For example, units or modules described herein, e.g., in a vehicle, drone or smartphone and/or in a server, may be, or may include, controller or processor **105**, memory **120** and executable code **125**.
(15) Storage system **130** may be or may include, for example, a hard disk drive, a universal serial bus (USB) device or other suitable removable and/or fixed storage unit. As shown, storage system **130** may include a database **131** that may include element representations **132** (collectively referred to hereinafter as element representations **132** or individually as element representation **132** merely for simplicity purposes).
(16) Objects in storage system **130**, e.g., element representations **132** and other objects in database **131** may be any suitable digital data structure or construct or computer data objects that enables

storing, retrieving and modifying values. For example, element representation objects **132** may be non-volatile memory segments, files, tables, entries or lists in database **131**, and may include a number of fields that can be set or cleared, a plurality of parameters for which values can be set, a plurality of entries that may be modified and so on. For example, a location, color, shape, unique identification or signature of an object, element or feature represented by an element representation **132** may be set, cleared or modified in the element representation **132** that may be associated with the actual element in a space, region or location.
(17) Attributes of an element (object or feature), e.g., included or described in an element representation **132**, may be, or may include any relevant information. For example, attributes of an element may be, or may include, a location of the element, an orientation of the element in space, a color of the element, a temperature of the element, a size of the element, a shape of the element or any other characteristics of the element.
(18) The phrase "element in a space" and term "element" as used herein may relate to any object, element or feature that can be sensed or identified by any sensor or modality. For example, an element (in a space) may be a house, a pole or a tree, e.g., sensed by a radar or a LIDAR system. In another example, an element can be a color of a wall, e.g., sensed by a camera. An element or feature can be, for example, a change of color, for example, if a first part of a wall is green and a second part of the wall is blue then the line separating the green part from the blue part may be a feature, e.g., sensed by a camera, as described. A color change feature may be recorded and used for determining a location as described. In yet another example, a temperature at a location can be a feature (e.g., sensed using an infrared (IR) sensor). Any object or phenomena in space may be an element as referred to herein. For the sake of simplicity, the term element will mainly be used herein, it will be understood that an element as referred to herein may mean, or relate to, any object or phenomena that can be sensed by any sensor.
(19) Additionally, or alternatively, the term "element" may refer herein to features and elements that may not necessarily be directly perceived by a human observer and/or by any type of sensor. For example, as elaborated herein (e.g., in relation to FIG. **2**), embodiments of the invention may include one or more AI engines (e.g., localization AI engine of FIG. **2**), adapted to extract one or more elements (referred to herein as "location indicative elements") that may that may be characteristic, or defining of a geographical location, but nevertheless may not be associated with any specific physical object and/or may not be perceivable by a human observer and or by any type of sensor.
(20) A unique identifier, reference or signature associated with an element, feature (or element representation **132**) may be unique within a specific instantiation of the invention, but not be unique when compared with the universe of numbers of data stored on all existing computer systems. For example, a unique identifier, reference or signature associated with an element may be computed based on attributes of the element, including a location of the element as well as other attributes, thus, for each real or actual element in space, a unique identifier may be calculated.
(21) Content may be loaded from storage system **130** into memory **120** where it may be processed by controller or processor **105**. For example, element representations **132** may be loaded into memory **120** and used for matching them with input from a sensor as further described herein.
(22) In some embodiments, some of the components shown in FIG. **1** may be omitted. For example, when computing device is installed in a vehicle, drone or smartphone, memory **120** may be a non-volatile memory having sufficient storage capacity and thus storage system **130** may not be required.
(23) I/O components **135** may be used for connecting (e.g., via included ports) or they may include: a mouse; a keyboard; a touch screen or pad or any suitable input device. I/O components **135** may be or may be used for connecting, to computing device **100**, devices or sensors such as a camera, a LIDAR, a radar, a heat sensing device, a microphone, an infra-red sensor or any other device or sensor. I/O components may include one or more screens, touchscreens, displays or monitors,

speakers and/or any other suitable output devices. Any applicable I/O components may be connected to computing device **100** as shown by I/O components **135**, for example, a wired or wireless network interface card (MC), a universal serial bus (USB) device or an external hard drive may be included in I/O components **135**.

(24) A system according to some embodiments of the invention may include components such as, but not limited to, a plurality of central processing units (CPU) or any other suitable multi-purpose or specific processors, controllers, microprocessors, microcontrollers, field programmable gate arrays (FPGAs), programmable logic devices (PLDs) or application-specific integrated circuits (ASIC). A system according to some embodiments of the invention may include a plurality of input units, a plurality of output units, a plurality of memory units, and a plurality of storage units. A system may additionally include other suitable hardware components and/or software components. In some embodiments, a system may include a server, for example, a web or other server computer, a network device, or any other suitable computing device.

(25) Some embodiments of the invention may automatically and/or autonomously, create a representation, view and/or mapping of a space based on input from a set of different sensing devices. A representation, view and/or mapping of a space (and elements, features or objects therein) may be created, e.g., in database **131**, by automatically creating element representations based on input from sensors of any type. A mapping between element representations and input from sensors may be automatically created. Accordingly, a system and method may automatically, based on input from a plurality of different sensors or modalities, create a representation, view and/or mapping of a space, e.g., such that objects or features in the space are usable for determining a location and/or orientation of a sensor. For example, after a mapping and representations are created, input from a sensor may be mapped to representations and the representations may be used to determine a location and/or orientation of the sensor.

(26) Some embodiments of the invention may provide a positioning solution based on matching features or objects captured by perception sensors to features or objects represented in a database. Some embodiments of the invention may match an element as seen or captured, from/by a first point of view or first modality with an element or feature as seen/captured from/by a second point of view or modality. Accordingly, unlike known location or positioning systems and methods, embodiments of the invention are not limited to specific sensors, modalities or technologies.

(27) For example, representations of elements, objects or features may be included in a database based on a top-view (e.g., satellite images) and these (representations of) elements, objects or features may be correlated or matched with elements, objects or features captured or seen using street-view images, e.g., captured by a camera in a vehicle. For example, elements in a top-view may be matched with elements in a ground surface view.

(28) Some embodiments of the invention may enable using different sensor technologies without the need to collect new data. Additionally, some embodiments of the invention may be adapted to determine a location and orientation in various, different scenarios and/or conditions, e.g., different road condition, weather and visibility conditions.

(29) As described, some embodiments of the invention may use auto encoding features to learn a correspondence or correlation between multiple instances of a scene, correspondence or correlation between multiple points of view and/or correspondence or correlation between representations of a scene by a plurality of modalities. The resulting system thereby learns localization features which are both discriminative or distinctive and invariant.

(30) Reference is made to FIG. **2**, an overview of a system **200** and flows according to some embodiments of the present invention. As shown, a system **200** may generally include at least one server **210** and at least one mobile device **230**. Server **210** may be, or may include one or more computing devices and/or modules that, for the sake of simplicity, will be referred to herein as server **210**. As shown, server **210** may include a dataset **215**, an analytics toolbox **216** and a database (DB) **217**. As further shown, server **210** may include, or be operatively connected to, a

localization artificial intelligence (AI) engine or unit **220**.
(31) Mobile device **230** may be a vehicle, a mobile computing device such as a smartphone or laptop, a drone or any other suitable computing device. As shown, mobile device **230** may include a DB **232** and an onboard localization and orientation (OLO) unit **233**.
(32) System **200** or components of system **200** may include components such as those shown in FIG. **1**. For example, server **210**, analytics toolbox **216** and AI unit **220** may be, or may include components of, computing device **100**. Mobile device **230** and OLO unit **233** may be, or may include components of computing device **100**. Although for the sake of clarity, a single mobile device **230** is shown in FIG. **2**, it will be understood that a system **200** may include a large number of mobile devices **230**. For example, server **210** may communicate with any (possibly very large) number of vehicles that may include the elements in mobile device **230**.
(33) According to some embodiments, system **200** may be scaled up by deploying a plurality of servers **210** that may be communicatively connected among themselves. Additionally, system **200** may be scaled up by including a large numbers of mobile devices **230**.
(34) As described herein, system **200** may be adapted to calculate, determine and/or provide (e.g., for a mobile device **230**), an exact location and orientation (e.g., of mobile device **230**) in space.
(35) As shown in FIG. **2**, input from one or more sensors included in, or connected to mobile device **230** may be received, as shown by sensor data **231**. According to some embodiments, sensor data **231** may be matched, in real-time or near-real time, with information produced by server **210** and provided to mobile device **230**. For example, information produced by server **210** and provided to mobile device **230** may be stored in DB **232**. Generally, any data or information included in database **131** or **217** may be communicated to, and stored in, DB **232**.
(36) For example and as described herein, based on an estimation of a location of a vehicle, some of element representations **132** may be downloaded to DB **232** of a specific mobile device **230**, such that DB **232** only stores element representations **132** that describe elements that are physically located near the specific mobile device **230**.
(37) The terms “real-time” (also referred to in the art as “real time”) and “near-real time” as referred to herein may relate to processing or handling of events at the rate or pace that the events occur or received (possibly defined by human perception). For example, a system according to embodiments of the invention may match input from sensor **311** (that may be a mobile sensor) with at least one element representation **132**, in real-time, e.g., within milliseconds or other very brief periods so that location and/or orientation of sensor **311** are made available or achieved virtually immediately.
(38) Dataset **215** may include information from any system or source. For example, dataset **215** may include information extracted from images, construction plans, road maps, blue prints, architectural plans and vector data of elements in a region or space. Dataset **215** may include information extracted from satellite images or view, street view, top view and a side view of elements in a region or space.
(39) AI unit **220** may use any method or system that includes artificial intelligence (AI) or machine learning (e.g., deep learning). Generally, machine learning (ML) or deep learning may relate to computerized systems or methods that perform a task without receiving or requiring specific instructions. For example, AI unit **220** may be a system that may be adapted to create at least one model **221** based on sample or training data. Analytics toolbox **216** may use the model **221** to make predictions or decisions without being explicitly programmed to perform the task of mapping input (e.g., from sensors **310** and **311**) to element representations **132** as described.
(40) A deep learning or AI model may use a neural network (NN). A NN may refer to an information processing paradigm that may include nodes, referred to as neurons, organized into layers, with links between the neurons. The links may transfer signals between neurons and may be associated with weights. A NN may be configured or trained for a specific task, e.g., pattern recognition or classification. Training a NN for the specific task may involve adjusting these

weights based on examples. Typically, the neurons and links within a NN are represented by mathematical constructs, such as activation functions and matrices of data elements and weights. A processor, e.g. CPUs or graphics processing units (GPUs), or a dedicated hardware device may perform the relevant calculations.

(41) According to some embodiments, localization AI engine **220** may be adapted to create and/or train one or more ML models **221**, such as deep-learning ML models, based on respective one or more sources of data. For example, localization AI engine **220** may receive (e.g., from dataset **215**) data that may include, for example, images, construction plans, road maps, blue prints, architectural plans, aerial images or any vector data. Additionally or alternatively, localization AI engine **220** may receive data originating from one or more sensors (e.g., a camera, a radar, a LIDAR sensor, an infrared sensor, and the like) such as element **310** of FIG. **3**. Localization AI engine **220** may be adapted to create or train a first ML model **221** corresponding to a first type of data (e.g., included in dataset **215**) and produce a second ML model **221** corresponding to a second type of data (e.g., data originating from the one or more sensors).

(42) According to some embodiments, localization AI engine **220 220** may be adapted to train at least one deep learning model **221**, based for example on data in dataset **215**. Model **221** may be adapted to identify or predict, as commonly referred to in the art, one or more features or elements that may be included in dataset **215**.

(43) For example, dataset **215** may include a plurality of data elements such as satellite and/or aerial photographs. AI engine **220** may produce a deep learning model **221** that may be or may include an autoencoder neural network. As known in the art, autoencoder neural networks may be trained so as to produce a compressed representation or an encoding of elements included in an input dataset. Thus, in a training stage, AI engine **220** may use a subset of the aerial photograph data elements as a training subset, to train model **221** so as to produce a representation **132** (e.g., compressed representation) or encoding of one or more elements (e.g., tree, houses, etc.) included in the training subset. In a subsequent inference stage, AI engine **220** may produce a compressed representation **132** or encoding of one or more elements (e.g., houses, poles, trees, etc.), that may be included in data elements (e.g., element **218**) of dataset **215** beyond the training subset. Additionally, or alternatively, in the inference stage, AI engine **220** may identify one or more elements or features (e.g., houses, poles, etc.) that may be included in data elements (e.g., element **218**) of dataset **215** beyond the training subset.

(44) In another example, dataset **215** may include a plurality of data elements that may originate from sensors (e.g., element **310** of FIG. **3**) that may be located at ground level, including for example, cameras, LIDAR sensors, radar sensors, etc. AI engine **220** may use a subset of the plurality of sensor (e.g., **310**) data elements as a training subset, to train model **221** so as to produce a representation **132** (e.g., a compressed representation) or encoding of one or more elements or features (e.g., trees, cars, houses, etc.) that may be included in data elements (e.g., element **231**, element **218**, data of sensor **310**) beyond the training subset.

(45) According to some embodiments of the invention, one or more (e.g., each) data element **218** of dataset **215** may be associated with a geographical position and/or orientation. For example, a data element **218** may be an aerial photograph, and may include, or may be associated with a location or position and/or an orientation at which the photograph was taken. In another example, a data element **218** may be an architectural plan (e.g., of a house, a constructed monument, and the like), and may be associated with a geographical location of a respective constructed element. In yet another example, a data element **218** of dataset **215** may originate from a sensor (e.g., an image of radar) and may include or be associated with a location at which the respective sensor data was obtained.

(46) According to some embodiments, localization AI engine **220** may be configured to produce at least one ML model **221**, adapted to predict one or more elements **320** or features that may be included in raw data **218** and may best correspond to geographical location. It may be appreciated

by a person skilled in the art that such elements may not necessarily correspond to actual, physical elements (e.g., a house, a pole) that may, for example, be perceived by a human eye. In some cases, such elements may be regarded as incoherent or amorphous by a human observer. Such elements are herein referred to as "location indicative" elements (e.g., element **325** of FIG. **3**). For example, embodiments of localization AI engine **220** may extract at least one location indicative element **325**, that may be or may include a combination (e.g., a numerical combination, such as a weighted sum) of a first number (e.g., representing the color of the sky) and a second number (e.g., representing a pattern of vegetation). Localization AI engine **220** may use the extracted at least one location indicative element **325** as a characteristic or an indicator of a geographical location (e.g., indicate whether an image has been taken in a mountainous environment or in a desert).

(47) Using the at least one model **221**, analytics toolbox **216** may create representations, descriptions and/or identifications of features and/or elements in a region or space. In other words, analytics toolbox **216** may perform inference of the at least one model **221** on a plurality of data elements.

(48) Pertaining to the example of aerial images, analytics toolbox **216** may perform inference of at least one model **221** on one or more aerial images originating, for example from dataset **215** (e.g., data element **218**), and may map or associate at least one representation, description and/or identification (e.g., a label of an identified object, such as a house) with a respective elements included in the aerial images.

(49) In another example, analytics toolbox **216** may receive a model **221**, adapted to analyze a specific type of input data, such as satellite images. In such embodiments, model **221** may have been trained on a first set of satellite images taken over one or more first locations, and adapted to identify objects (e.g., road signs, buildings) in the images. Analytics toolbox **216** may infer model **221** on one or more satellite images (e.g., data elements **218**) of dataset **215**, taken over one or more second locations, to identify objects in the one or more second locations. As elaborated herein, data elements **218** may be associated with specific geographical locations. Therefore, analytics toolbox **216** may produce one or more element representations **132** corresponding to the one or more identified objects (e.g., road signs, buildings, etc.), and the one or more element representations **132** may be associated with the specific geographical locations. Said element representations **132** associated with specific geographical locations may herein be referred to as road codes (e.g., element **250**).

(50) In yet another example, localization AI engine **220** may produce an ML model **221** that may be adapted to receive at least one input data element of a specific type or modality, such as a specific type of sensor (e.g., element **310** of FIG. **3**). ML model **221** may be adapted to extract or predict from the at least one received data element one or more location indicative elements **320**, as elaborated herein. Analytics toolbox **216** may be configured to receive at least one data element, such as data element **218** from dataset **215**, that may correspond to the same specific type or modality. Analytics toolbox **216** may use ML model **221** to extract at least one location indicative element **325** that may be included in input data element **218**, and may be a characteristic of a geographical location. As elaborated herein, data element **218** may be associated with a geographical location or position. Therefore, analytics toolbox **216** may create an element representation **132** that may also be associated with the same geographical position. In other words, analysis toolbox **216** may produce, from the incoming data element **218** at least one road code **250** that may be associated with an location indicative element **325**, based on ML module **221**.

(51) The term "mapping" may refer herein to an association of a label or a representation of an element **132** with one or more data elements (e.g., one or more images and/or one or more portions thereof) that may be, for example, included in dataset **215** and/or obtained from a sensor (e.g., element **310** of FIG. **3**).

(52) According to some embodiments, analytics toolbox **216** may update DB **217** so as to include said mapping or association between element representations and corresponding input data (e.g.,

218 and/or data of sensor **310**).
(53) In some embodiments, one or more representations, descriptions and/or identifications of features and/or elements may be provided to mobile device **230**. For example, and as shown in FIG. **2**, mobile device **230** may receive one or more road codes **250**, and may be stored in DB **232**.
(54) According to some embodiments, OLO unit **233** may extract road-codes **250** and/or other representations, descriptions and/or identifications of features and/or elements from DB **232**, match or correlate them with sensor data **231** (produced by sensors in mobile device **230**) and, based on the matching, determine its location and/or orientation **260** in relation to the relevant representations. Location and orientation **260** may be presented to a user, e.g., using a screen, speakers and the like.
(55) As shown by the arrow connecting OLO unit **233** and analytics toolbox **216**, OLO unit **233** may update analytics toolbox **216**, e.g., to remove an element that is no longer present in a region, add an element that is missing in DB **217**, modify a description of an object and the like.
(56) Reference is additionally made to FIG. **3** which shows components and flow according to some embodiments of the invention. As shown, an element representation **132** and input from a first sensor **310** may be used to create a mapping **330** between the input and the representation. As further shown, using the mapping **330**, input from a second sensor **311** may be matched with the element description **132**. Sensors **310** and **311** may for example be mobile sensors, e.g., installed in a car, aircraft, ship or bike or hand carried by a user, e.g., included in a smartphone.
(57) Additionally, or alternatively, at least one sensor **310** may be adapted to obtain at least one input data element that may pertain to at least one location indicative element **325**. As elaborated herein, location indicative element **325** may not pertain to any specific physical object (such as house **320** or pole **323**), but may be a produce of ML model **221**. According to some embodiments, location indicative element **325** may be unintelligible or incoherent to a human observer, but may be recognized by localization AI engine **220** as indicative of geographical locations. In such embodiments, analytics toolbox **216** may perform mapping between a representation **132** of location indicative element **325**, and the input data of the at least one sensor **310**.
(58) Mapping **330** may be, or may include any information or logic that, provided with input from a sensor, produces at least one value, indication, pointer or reference that enables identifying, locating or extracting, at least one element representation **132** from a database. Otherwise described, mapping **330** may be, or may be viewed as, a conversion unit, data or logic that converts input from a sensor to an element representation **132** object.
(59) For example, a first sensor (e.g., element **311**, such as a radar) may be adapted to sense an object (e.g., element **323**, such as a pole). First sensor **311** may be adapted to subsequently output a signal or a data element corresponding to the sensed object. In the example of the pole, first sensor **311** (e.g., the radar) may produce one or more or more signals or data elements that correspond to one or more (e.g., a set of) values or readings that represent a frequency, an amplitude or other attributes of RF radiation or electromagnetic waves, reflected from pole **323**. The set of values may be mapped to, or associated with, an element representation object **132**, accordingly, when the set of values is received at a later stage (e.g., by another radar sensor **311**), the mapping or association can be used to map the values to the element representation **132** object of pole **323**.
(60) As described, mapping **330** may generate a signature based on a set of values received from a sensor as described and the signature may be used for finding the correct element representation **132** object, the signature may be used as a key for finding an element representation **132** object in a database, e.g., in DB **232**. For example, OLO **233** may examine sensor data **231** (input from a sensor), calculate or produce a signature based on the input and use the signature to find an element representation **132** object in DB **232**. Once an element representation **132** object is found, attributes such as the location of the element may be extracted from the element representation **132** object.
(61) For example, an element in space **340** may be a corner of a house **320**, a house **321**, a pole or tree **323** or a change of color **322**, e.g., element or feature **322** may be a line separating a white part

of a wall from a black part. Elements, objects or features **320**, **321**, **322**, **323** and **325** may be represented in database **131** using element representations **132**. Element representations **132** may include representations or encoding of elements that may originate from AI model **221**. For example, AI model **221** may be or may include an autoencoder model, having a bottleneck hidden layer, which may be adapted to produce a representation (e.g., a compressed representation) or encoding of elements included in an input data set. **132**.

(62) Generally, an element representation **132** may include any metadata related to an element, object or feature. The terms element, object and features as referred to herein may mean the same thing and may be used interchangeably herein. For example, an element representation **132** may include a location (e.g., using coordinate values as known in the art), a color, a size, a shape and the like. For example, an element representation **132** of house **321** may include the house's location, an element representation **132** of pole **323** may include the pole's height and an element representation **132** of color change **322** may include the location and length of a line that separates two colors on a wall as well as an indication of the two colors.

(63) Generally, a coordinate system enables using values to represent a location and/or orientation in a space. For example, coordinates in a Cartesian coordinate system are tuples that represent a location of a point in space. A coordinate system may be standard or global, e.g., known and used by various systems and methods or it can be a relative one. For example, a relative coordinate system may represent a location of an element in space with respect to sensor **311**, e.g., a relative coordinate system may be used to indicate the location of house corner **320** with respect to sensor **311**, e.g., the origin (e.g., the point 0,0,0) of a relative coordinate system may be sensor **311** such that the relative locations of pole **323** and house **321** are represented with respect to the location of sensor **311**.

(64) It is noted that the point of view of sensor **310** may be different from the point of view of sensor **311**, for example, sensor **310** may be installed on a rod extending from a roof of a car traveling from north to south near house **321** and sensor **311** may be placed on a bumper of a car traveling in the opposite direction near house **321**, as described, an element representation **132** of an element in a database may be based on a point of view (or modality or sensor) that is different from the point of view, sensor type or modality of sensors **310** and **311**, accordingly, embodiments of the invention enable matching representations of elements that are derived using different modalities, input types or sensors types. Accordingly, sensors (e.g., sensors **310** and **311**) may be installed in different vehicles or in different stationary objects.

(65) In some embodiments, a method of determining location and orientation of a sensor may include storing or including, in a database, a representation of an element in a first space and creating a mapping or correlation between the representation and input from a first perception sensor (e.g., sensor **310**). For example, based on a satellite image, a representation **132** of house **321** seen in the satellite image, may be included as an element representation **132** in database **217** (e.g., element **131** of FIG. **1**), thus forming a representation of an element in a first space. Subsequently, input from a camera (first perception sensor, e.g., sensor **310**), e.g., in a vehicle traveling by house **321**, may be received (forming the representation and input from a first perception sensor). An embodiment may examine the image, identify house **321** therein and define a mapping (e.g., mapping **330**) between the image representation of house **321** in the image and the element representation **132** in database **217** (e.g., **131**) which describes house **321**. It will be noted that a sensor that is a conventional (visible light) camera and that produces an image is just one of the types of applicable sensors. For example, sensor **310** may be a LIDAR device, a radar, a heat sensing device, a microphone, an infra-red (IR) sensor and the like.

(66) Processing and operations described herein may be performed by a server (e.g., server **210**) and/or by a unit in a vehicle (e.g., OLO **233**) or a server and a unit in a vehicle may collaborate in performing methods described herein. For example, server **210** may include in a database a representation of an element in a first space, e.g., in the form of an element representation object

132 and the server may further create a mapping **330** between the representation and input from a sensor (e.g., **310**) and provide the mapping to OLO **233** which may use the mapping and input from a second perception sensor (e.g., **311**) to identify one or more elements in a database (e.g., in DB **232**) and use attributes of elements identified in the database to determine a location and/or orientation of mobile device (e.g., vehicle) **230**.
(67) It is noted that, e.g., in the above example involving a camera, once a mapping **330** is created as described, any image of house **321** captured by practically any camera or image acquisition device may be mapped to the element representation **132** of house **321** in database **131** (e.g., element **217** and/or element **232**). For example, one or more attributes of house **321** such as size, shape, height, color, contours and the like may be taken into account when creating or defining mapping **330**, such that any image of house **321** may be mapped, using mapping **330**, to the correct element representation **132** in database **131** (e.g., element **217** and/or element **232**), that is, to the specific element representation **132** that describes house **321**.
(68) It may be appreciated that although the example of a house (e.g., elements **320**, **321**) is used ubiquitously herein, similar logic may be applied to any other physical feature or element (e.g., color change **322**, pole **323**), or on non-physical elements, such as location indicative element **325**.
(69) In some embodiments, a method of determining location and orientation of a sensor may include using a mapping and input from a second perception sensor, in a second space, to locate or identify one or more elements in the database. For example, using mapping **330**, input from sensor **311** (second sensor) may be used to locate or identify house **321** in DB **232**. For example, an image of house **321**, as captured by sensor **311** (that may be a camera or other sensing device) may be processed using mapping **330** to produce a result that may be used to locate element representation **132** that describes house **321**, in database **131** or in DB **232**.
(70) In some embodiments, a method of determining location and/or orientation of a sensor may include using attributes of identified (e.g., in a database) one or more elements to determine a location and orientation of the second perception sensor in the second space. For example, provided with an image of house **321** captured by sensor **311** (the second sensor) and using mapping **330**, OLO unit **233** may produce a reference, key or signature that can be used for extracting, from DB **232**, the element representation **132** object that describes house **321**.
(71) As described, the element representation **132** object that describes house **321** may include any metadata or attribute, e.g., an attribute or metadata in the element representation **132** object may be the exact location of house **321** in the form of coordinates values in a global or known coordinate system. Accordingly, exact locations of elements **320**, **321**, **322**, **323** and **325** may be obtained by OLO unit **233** thus, e.g., using triangulation, OLO unit **233** may accurately determine its location and/or orientation.
(72) In some embodiments, a method of determining location and orientation of a sensor may include creating a first mapping according to a first condition and creating a second mapping according to a second condition and selecting to use one of the first and second mappings according to a condition.
(73) For example, sensor **310** may be used to sense house **321** in daylight and then at night and two mappings **330** may be created, one to be used during the day and another for night time. Similarly, mappings **330** may be created for summer, winter, storm, specific dates, time of day and so on. Similarly, a first mapping **330** of input from sensor **311** when sensing pole **323** on a rainy day to an element representation **132** object may be created and a second mapping, to the same element representation **132** may be created for input from sensor **311** when sensing pole **323** on a sunny day. Accordingly, a mapping may be based on a condition, time of day, date and so on. As described, based on a condition, a mapping may be used, e.g., if pole **323** is sensed or “seen” by sensor **311** during day time then the day time mapping may be used and, if the pole is sensed by sensor **311** during the night (thus input from sensor **311** may be different) then a night mapping may be selected. Accordingly, a mapping may be created and dynamically selected based on a

condition. For example, OLO **233** may select a first mapping if the weather is fine and a second mapping in a storm, e.g., OLO **233** may select a first mapping in a first condition (e.g., cold weather or rainy day) and select a second mapping in a second condition (e.g., warm weather or sunny day).

(74) In some embodiments, elements (e.g., element representations **132**) may be classified. For example, element representations **132** may be classified according to a suitability for sensor type (e.g., a first class of elements may be suitable for use with a camera and a second class of elements may be suitable for use with a LIDAR sensor or device). Element representations **132** may be classified according to a condition, e.g., elements best detected at night may be associated with a first class and elements best detected in sunlight may be associated with a second class. Similarly, classification of elements may be according to weather conditions, location (e.g., elements in a tunnel may be classified differently from elements in a farm and so on. Any aspect or condition may be used for classifying elements. Accordingly, an embodiment can automatically select a set of elements to be used for determining location and/or orientation based on any condition or aspect. For example, if it rains when location of sensor **311** is to be determined then a set of elements classified as best for poor visibility may be selected and used as described herein for determining the location and/or orientation of sensor **311**.

(75) In some embodiments, a representation **132** of an element in database **131** may be based on, or related to, a first point of view and input from a sensor for which location and orientation is determined is related to a second, different point of view.

(76) For example, a first point of view may be used for creating mapping **330** and a second, different point of view may be used for determining location and orientation of a sensor. For example, the point of view of sensor **310**, when sensing house **321** to create mapping **330** as described, may be different from the point of view of sensor **311** when its location and orientation are determined based on identifying house **321** as described.

(77) In some embodiments, a plurality of mappings **330** may be created for a respective plurality of sensor types or modalities. In some embodiments, a method of determining location and orientation of a sensor may include selecting one of a plurality of mappings **330** based on the type of the sensor.

(78) For example, in a first case, mobile device **230** may inform server **210** that it includes a camera (sensor type), in response, server **210** may search database **131** or DB **217** for elements or features that can be sensed by a camera, e.g., colors, road signs and the like. Next, server **210** may send to mobile device **230** (e.g., as road codes element **250**) the mappings for the selected elements, e.g., the mapping **330** of color change **322** may be downloaded to mobile device **230** since it's best suited for sensing by a camera. In a second case, mobile device **230** informs server **210** that it includes a LIDAR system, in this case, server **210** may send mapping **330** for pole **323** but avoid sending the mapping for feature **322** since a change of color on a surface cannot be accurately sensed by a LIDAR.

(79) Accordingly, an embodiment may create a plurality of mappings **330** for a respective plurality of sensor types and automatically or dynamically select a mapping to be used based on the type of sensor for which location and/or orientation are determined, an embodiment may further dynamically or automatically select a subset of elements in a database, wherein the subset selected based on the type of the second sensor, that is, both mappings and element representations **132** may be selected such that the set of element representations **132** and mappings **330** downloaded to DB **232** is the optimal set for the sensor used. Of course, if mobile device includes a number of different sensors or different types of sensors (e.g., camera and LIDAR) then element representations **132** and mappings **330** for both camera and LIDAR may be automatically selected and downloaded to mobile device **230** as described.

(80) Accordingly, embodiments of the invention may select, from a large set, a subset of element representations **132** and mappings **330** based on a location of a sensor and based on attributes of the

sensor, e.g., the sensor type and/or suitability of the sensor to sense different or specific elements in the vicinity of the sensor. As described, the set may also be selected based on a condition.
(81) In some embodiments, an element representation **132** in one or more of database **131**, DB **217** and DB **232** may be updated based on input from a sensor. For example, starting from a satellite image, a plurality of circles may be identified, and their respective diameters and locations may be included in their respective element representations **132**. Next, an image may reveal that the circles are the top surfaces of poles, in such case, the element representations **132** already in database **131** may be updated to reflect the fact that the elements or poles. Any metadata related to an element or feature may be updated, or modified as described. For example, in the above example of poles, updating the element representations **132** may include indicating which sensors can sense them, in what conditions the elements can be sensed and so on.
(82) In some embodiments, e.g., if input from sensor **311** cannot be mapped to any element representation **132**, a new element representation **132** may be created in database **131**. For example, if pole **323** was recently placed by a road then input from the first sensor **310** that passes by pole **323** may cause controller or processor **105** to add a new element representation **132** to database **131**. As described, input from additional sensors passing by pole **323** may be used to verify that pole **323** is present and/or map the element representation **132** of pole **323** to other sensor types as described.
(83) Element representations **132** may be removed from database **131**. For example, after pole **323** is removed from the road side, sensor **311** does not sense it when passing where pole **323** was placed, by comparing input from sensor **311** to data in DB **232**, controller or processor **105** may identify that pole **323** is no longer sensed and may remove the relevant element representation **132** from all databases in system **200**. Accordingly, embodiments of the invention continuously and autonomously update a representation of space such that it truly reflects the current state of the space.
(84) The advantages of automatically and autonomously creating digital representations of elements, objects or features as described may be appreciated by those having ordinary skill in the art. For example, assuming a new sensor, e.g., an IR sensor, is introduced to the field or market, e.g., input from such a type of IR sensor has never been received by system **200**. When first receiving input from the new IR sensor, system **200** may automatically create element representations **132** based on the input. An embodiment may then create a mapping between features, elements or objects sensed by IR sensors and element representations **132**, e.g., during a training session. An embodiment may associate or map element representations **132** created based in IR sensors with other elements in a database such that information related to a newly introduced sensor are identified and characterized using input from other sensors and/or input from other sources, e.g., maps and the like. As described, element representations **132** added may be used to determine a location and/or orientation of IR sensors. Accordingly, a system and method need not be configured or modified in order to adapt to new sensor types, rather, systems and methods of the invention can automatically adapt to newly introduced sensors or modalities.
(85) A versatility with respect to sensor types or modalities provided by embodiments of the invention cannot be provided by known systems and methods. The versatility with respect to sensor types or modalities is an improvement current or known systems and methods which are restricted to specific sensors or modalities. For example, sensor **311** may be a microphone and the sound of a power generator in a factory (e.g., intensity and frequency) may be used for creating an element representation **132** object that may be used to determine location and/or orientation as described. Similarly, any phenomena that can be sensed by any sensing device may be represented by an element representation **132** in database **131** and used as described, e.g., element representations **132** objects may be created and/or updated based on optical data, acoustical data and radio frequency data.
(86) By supporting any modality or sensor type as described, embodiments of the invention

improve the field of location sensing or determination by providing scalability and configurability that cannot be provided by known systems or methods. For example, a user who just added a LIDAR system to his car can immediately receive location and/or orientation information based on the newly added LIDAR system and based on element representations **132** that are relevant to LIDAR systems, e.g., based on element representations **132** and a mapping that were previously created or updated based on input from LIDAR systems. A system and method may be trained to support any sensor type or modality as described. It will be noted that once a system is trained to use a specific sensor or modality, it may be used to determine a location and/or orientation of a sensor even in places or spaces where no input from the specific sensor type was received, e.g., using a mapping as described, any object in any space may be identified as described.

(87) Generally, an element or feature as referred to herein may be any object, element or feature that can be sensed by any sensor or modality, e.g., an element or feature may be sensed by, or based on, a light signal, an audio signal, temperature or heat, moisture level, electromagnetic waves or any other phenomena that can be sensed. For example, moisture near a carwash place or water park may be sensed, heat from a white wall (as oppose to heat from a black wall) may be sensed and so on. For example, a first element representation **132** may describe an object sensed by sonar, a second element representation **132** may describe an object sensed based on heat (e.g., using IR) and so on.

(88) Element representations **132** may be created based on any source, modality or type of data, e.g., they may be created based on an images, construction plans, road maps, blue prints, architectural plans, aerial images or any vector data. For example, element representations **132** may be created, e.g., by server **210**, based on a satellite image or view, a street view footage or images, a top view and a side view all of which may be obtained by server **210** from any source, e.g., the internet, authorities and so on. Accordingly, any modality (and, as described, any data from any point of view) may be used for creating element representations **132**. For example, an element representation **132** in database **131**, related to an object in space may be created based on a mark in a map, may then be updated based on an image of the object, then updated based on input from a LIDAR device and so on. As described, element representations **132** in database **131** may be found, identified and/or retrieved based on input from a sensor (of any type) using a mapping as described.

(89) In some embodiments, a global or common coordinate system may be used. For example, as described, an element representation **132** may include location information in the form of coordinates in a first coordinate system, in case a second, different coordinate system is used by mobile device **230**, an embodiment may convert or transform location information such that it is presented to a user according to a preferred or selected coordinate system. In some embodiments, an element representation **132** may include location information (e.g., coordinates) related to one or more coordinate systems, e.g., related to a global coordinate system and a number of different local coordinate systems, in other embodiments, a conversion logic may be used to translate or transform location information as required.

(90) In some embodiments, the information in a database that is matched with input from a sensor as described is selected based on received or calculated location information. In some embodiments, an estimation of a location of the sensor in a space, area or region is determined based on location information related to the sensor. For example, based on GPS data, a cellular tower's location information or Wi-Fi access points identifications, at list the area where sensor **311** is in may be determined and server **210** may select to download to DB **232** elements in the area. Accordingly, only relevant elements in a space or region may need to be stored by mobile device **230** thus eliminating the need of mobile device **230** to store, and search through, large amounts of data.

(91) In some embodiments, mapping **330** may map or correlate any input from a sensor to an element representation **132**, e.g., raw data received from sensor **311** when sensing corner **320** may be processed using mapping **330** to produce a signature, pointer or reference that can be used to

identify the element representation **132** that describes corner **320**.

(92) In some embodiments, data received from a plurality of sources, e.g., a plurality of sensor types, may be encoded to produce a unified format representation of the data and the unified format representation may be included in the database, e.g., in one or more element representations **132**. By encoding different data types coming from different sensor types into a unified format representation, an embodiment can represent a single element or feature in space based on input from a plurality of different modalities. For example, an element representation **132** may be created based on a map (first source) and a satellite image (first modality) and based on input from a mobile sensor such as a mobile camera (second modality), such creation of an element representation **132** based on input from a plurality of sources, sensor types and modalities may be done using a unified format representation. In some embodiments, a unit (not shown) connected to a sensor in mobile device **230** may convert raw data received, e.g., from sensor **311**, to a unified format representation (that may be included in, or associated with, element representations **132**, e.g., in the form of metadata as described), thus, efficiency of a search in DB **232** for an element representation **132** is improved and speed of operations described herein is increased.

(93) Adapted to use input from any type of sensor, embodiments of the invention improve the technology which is currently confined to a limited, preconfigured set of sensors or modalities, embodiment of the invention provide a practical application of a process for determining a location and/or orientation of a sensor (or vehicle or other entity that includes the sensor), for example, a practical application of such process includes OLO **233**, DB **232** and at least one sensor **311** installed in a car such that a system can provide the driver with his or her location based on input from a variety of sensors that may be of any type or modality.

(94) In some embodiments machine learning may be used to match input from a sensor with elements. For example, an iterative learning process may include examining a set of inputs, from one or more sensors, related to an object and generating a model that, given input of a sensor sensing the object maps the input to an element representation **132** that is associated with the object.

(95) Reference is made to FIG. **4**, a flowchart of a method according to illustrative embodiments of the present invention. As shown by block **410**, a representation of an element may be included in a database. For example, a plurality of element representations **132** may be included in database **131**. As shown by block **415**, a mapping between the representation and input from a first sensor may be created. For example, based on input from sensor **310**, when sensing pole **323**, an embodiment may create a mapping **330** such that, using the mapping, input from a sensor near pole **323** is mapped to an element representation **132** that describes pole **323**.

(96) As shown by block **420**, the mapping may be used to map input from a second sensor, in a second space, to identify one or more elements in a database. It will be noted that the first and second sensors may be of the same type, or they may be of different types. The first and second spaces may be the same space in different times or they may be different spaces. For example, if a mapping enables identifying pole **323** in DB **232** then similar or identical poles (in another place or space) may be identified using the mapping created for pole **323**. As shown by block **425**, using attributes of the identified one or more elements, a location and/or orientation of a sensor may be calculated and/or determined. For example, by identifying a number of elements around sensor **311** as described, the exact locations of these elements may be known (e.g., since it may be included in their respective element representation **132** objects) thus the location of sensor **311** may be determined, either globally or with respect to the identified elements. For example, a location provided to a user by an embodiment may be in the form of coordinate values and/or in the form of an indication of the location on a map, e.g., as done by navigation systems. An orientation provided to a user may be in the form of an image or avatar (e.g., of a car) where the image or avatar is shown, on a map, such that the orientation (e.g., direction of movement) are clearly shown.

(97) Some embodiments may continuously update, add and/or remove element representations **132**,

e.g., based on iteratively receiving input from sensors and updating metadata in, or of, element representations **132** objects or other information in database **131**. For example, recorded inputs from a plurality of sensors **311** that travel near elements, objects or features **320**, **321**, **322**, **323**, and **325** over a possibly long time period, may be anchored or fixed in using a common coordinate system such that accuracy, e.g., of the location of element **320** (as included in metadata of the relevant element representation **132**) verified and/or is increased. For example, using the locations of sensors **311** when they sense feature **322**, a set of inputs from these sensors may be correlated or brought into a common representation, e.g., a common coordinate system.

(98) Accordingly, the set of inputs may be used for increasing accuracy of location and/or orientation of elements, thus, accuracy of location and/or orientation of a sensor provided by embodiments of the invention is increased.

(99) In some embodiments, location and/or orientation of a 3D space may be determined by obtaining a 3D digital representation of a space using input from a mobile sensor; and determining a location and/or orientation of the space by correlating the 3D representation with a top view representation of the space. For example, sensor **311** may be a video or other camera used for capturing a set of images of a street (e.g., street view footage). The set of images may be used to create a 3D representation of a space that includes or contains the street. The 3D representation may then be correlated with a top view of the street, e.g., a satellite image.

(100) In some embodiments, correlating a 3D representation with a top view may include projecting the 3D representation on the top view. For example, the 3D representation of the street may be projected on, aligned (or registered) with a satellite image or a road or other map in database **131**. A map or image in database **131** with which a 3D representation may be correlated as described may include, or may be associated with, metadata related to elements, features or objects, e.g., metadata included in element representations **132**. For example, the location (e.g., in the form of coordinate values in a global coordinate system) of elements in a map may be known. Accordingly, by correlating elements in a 3D representation with a map or other data in database **131**, an embodiment may determine the exact location and/or orientation of the space in a coordinate system. For example, by projecting the 3D representation of the street on a map and aligning the 3D projection with the map, the location and orientation of the street may be determined.

(101) As described, an embodiment may determine a location and/or orientation of a sensor in a space. In some embodiments, having determined the exact location and/or orientation of a space in a coordinate system as described, the exact location and/or orientation of a sensor in the space may be determined (e.g., by identifying elements in the space as described). For example, if the relative location of sensor **311** with respect to a number of elements in a space (e.g., house **321**, pole **323** etc.) is determined as described and the location and orientation of the space are determined as described (e.g., by registering or projecting a 3D object on a map) then an embodiment may readily determine the location and/or orientation of sensor **311** in any coordinate system. Accordingly, to determine a location and/or orientation of a sensor, an embodiment may capture images around the sensor, generate a 3D representation of the space surrounding the sensor, determine the exact location and orientation of the space in a global coordinate system and, based on the (possibly relative) location and/or orientation of the sensor in the space, determine the location and/or orientation of the sensor in the global coordinate system. For example, while a vehicle is traveling through a street, an embodiment may capture images (or video footage) in the street, create a 3D representation of the street, determine the exact location and/or orientation of the street (e.g., in a global or known reference such as a map) and, based on the location and/or orientation of the vehicle in the street, determine the exact location and/or orientation of the vehicle with respect to a map or any global or common reference of choice.

(102) In some embodiments, a 3D digital representation may be created based on a set of images and based on a relative location of at least one element in at least some of the images. For example,

devices or systems such as an odometer, Inertial Measurement Unit (IMU) simultaneous localization and mapping (SLAM) system may be used to determine a relative point of view (or location and/or orientation) for some or even each of a set of images obtained as a camera or other sensor is moving through a space, e.g., traveling in a street. Using the relative point of view of a set of images, a 3D digital representation of a space may be created.

(103) Although, for the sake of clarity and simplicity, a camera and images are mainly discussed herein, it will be understood that input from any sensor or modality may be used for determining a location and orientation of a space by correlating a 3D representation with a top view representation of the space. For example, instead of, or in addition to, a camera as described, a LIDAR or imaging radar system may be used for creating a 3D representation.

(104) In some embodiments, a top view representation of a space includes at least one of: an aerial image, a satellite image, a structural map, a road map and a DEM. Any information or data may be used to create a top view digital representation without departing from the scope of the invention.

(105) In some embodiments, a portion of a top view representation may be selected based on location information of the space. For example, using GPS or other means, a location or an estimation of a location of sensor **311** may be determined and a portion of a satellite image covering the area of sensor **311** may be selected for the projection of a 3D digital representation as described, such that speed and efficiency are increased and/or amount of required memory or computational resources is reduced.

(106) As described, embodiments of the invention provide a number of advantages over known or current systems and methods. For example, a system and method may autonomously and/or automatically identify (and add to a database representations of) elements, objects or features. Moreover, a system and methods may identify (and add to a database representations of) elements, objects or features that would not be naturally, or instinctively, identified by a human. For example, to guide a friend coming to visit, one might tell the friend “make a left right after the big oak tree” since a tree is something readily identified by humans. In contrast, a system and method may identify a part of the oak's trunk as an element and add a representation of the identified part of the trunk to DB **217**. For example, a part of a tree trunk identified by a LIDAR system may be an element represented by an element representation **132** since it is readily and/or accurately identified by a LIDAR system while a camera or human would have a difficulty identifying such element. For example, a human would not be able to identify a tree with a trunk that is two feet in diameter, however, an embodiment may identify the trunk's diameter with great precision, represent the trunk in DB **217** and consequently use the tree trunk for determining a location as described.

(107) Since embodiments of the invention may identify, define (and represent in a database, e.g., in element representations **132**) elements, objects or features in space based on any sensor or modality, embodiments of the invention can identify (and represent in a database and use to determine a location as described) elements, objects or features that cannot be identified by known systems and methods nor by humans. Otherwise described, while known systems and methods (and humans) rely on objects or elements such as houses or road signs, embodiments of the invention create a view of a space based on the way that any sensor or modality “sees” or “senses” space. To illustrate, the sound of a power generator picked up by a microphone cannot be used by any known system or method to determine a location, however, for some embodiments of the invention, the sound of the generator (e.g., its audio characteristics) is a perfectly legitimate feature or element that may be represented by an element representation **132** and used for determining a location as described. By creating a view of space based on how it is sensed by a variety of sensors, embodiments of the invention provide an innovative way of representing space and navigating therein.

(108) Using machine learning or deep learning, embodiments of the invention may create a representation of an element, object or feature based on input from a first type of sensor or modality and then identify the element, object or feature based on input from a second, different

type of sensor or modality. For example, using machine or deep learning, an embodiment may know, or predict the input that a LIDAR system will provide when sensing a tree trunk as described. Accordingly, provided with an image of a tree trunk, an embodiment may accurately predict the input that a LIDAR system will provide when traveling near the tree. For example, based on an image from a camera, DB **217** may include an element representation **132** describing a pole **323**, based on a mapping **330** created as described, an embodiment may know, or predict, the input that a LIDAR system will provide when sensing (or traveling near) pole **323**.

(109) Accordingly, embodiments of the invention can create a representation of an element in a space based on a first type of sensor and then identify the element based on input from a second, different type of sensor without having to collect additional data, e.g., an image of pole **323** and a mapping **330** that translate elements in an image to input from a LIDAR system may suffice for OLO **233** in order to identify pole **323** based on input from a LIDAR device even though this may be the first time input from a LIDAR that senses pole **323** is ever received. Otherwise described, an embodiment may create an element representation **132** for an object in space based on input from a first sensor type (e.g., a camera) and then identify the object based on input from a second, different sensor type even if the object has never before been sensed by a sensor of the second type.

(110) Mapping **330** may be, may be included in, or may be created based on, a model created using machine or deep learning or NN. Mapping **330** may create, define or include a mapping or link between two or more sensor types or sources of information with respect to objects, elements or features in space. Machine learning may be used, by embodiments of the invention in order to understand the connection between two or more sources of information. For example, machine learning may be used to associate input from a LIDAR system to an element representation **132** where the element representation **132** was created based on an image. For example, provided with input from a LIDAR when sensing houses **321**, a system may train a model usable for translating input from LIDAR sensors such that the input is mapped to element representations related to houses **321**. Otherwise described, a model may link input from LIDAR systems to input from a camera in a way that enables creating element representations **132** of houses based on images and then determining that an element sensed by a LIDAR is one that is represented by an element representation **132** created based on an image. Of course, LIDAR and camera are only some examples of modalities brought here as examples

(111) For example, mapping **330** may map input from a camera to input from a LIDAR system such that based on an image of objects such as pole **323** and features such as color change **322**, input from a LIDAR system can be mapped to these objects (e.g. the input may be mapped to an element representation **132**), thus, provided with an element representation **132** of pole **323** created based on an image of pole **323** (and based on data in DB **232**), OLO **233** can determine presence of pole **323** based on input from a LIDAR system. Generally, element representations **132** may describe elements or objects which are fixed in time and space and using mapping **330** as described, these elements may be identified using input from any sensor. As described, element representations **132** may be automatically created based on input from any sensors and, as described, element representations **132** may describe elements or features that are not necessarily ones that are native (or even visible) to a human or camera, rather, element representations **132** may be created based on input from any sensor or modality.

(112) It will be understood that once a mapping **330** was created for a specific modality or sensor type with respect to an object or object type, the object or object type (or instances of the object type) may be identified in any place, location or space. For example, a system may be trained to identify poles **323** based on input from a LIDAR sensing poles **323** in a first town or region (first location) and, once the training is complete, input a LIDAR system in another city or region (second location) may be identified as related to poles **323** or it may be used to determine presence of poles **323**. For example, at a first step, training of a model (or creating a mapping) may be done such that input from a LIDAR system can be readily mapped to element representations **132** of

poles, e.g., based on input from a LIDAR system in a first city, region or part of the world and based on knowledge of presence of poles in the region. At a next step, when mobile device **230** is located (or traveling) in another city or region, element representations **132** describing objects in the second city or region are downloaded to DB **232** and the model or mapping are also downloaded to DB **232**. Next, using input from a LIDAR device in mobile device **230** and the model or mapping, OLO **233** may identify poles in the second city, e.g., identify, in DB **232**, element representations **132** of nearby poles. It will be noted that OLO **233** needs not know that the identified objects are poles, all OLO **233** needs to determine is that input from a LIDAR system corresponds to an element representation **132** in DB **232**, as described, this correspondence may be achieved based on a model that maps input from the LIDAR system to an element representation **132**. For example, when mobile device **230** is a vehicle driven in Paris then server **210** may download to DB **232** element representations **132** that describe (or that are related to) elements, features or objects in Paris, using a model or mapping **330** as described, OLO **233** may map input from a LIDAR or any other sensor type to element representations **132** in DB **232** even though the mapping or model were created based on input from a LIDAR system located in London, of course, at least some element representations **132** related to objects in Paris need to be included in DB **217** (and downloaded to DB **232**) such that they can be found by OLO **233** when traveling in Paris. For example, server **210** may update DB **217** and/or dataset **215** based on input from sensors from anywhere in the world such that element representations **132** describing objects anywhere in the world are included in DB **217** and/or dataset **215** and thus the relevant element representations **132** can be downloaded to a vehicle (mobile device **230**) based on a region or area where mobile device **230** is located.

(113) Accordingly and as described, some embodiments of the invention may include in a database representations of elements in a region or space based on input from a first sensor type or modality (e.g., a camera) and subsequently map input from a second, different sensor type or modality (e.g., a LIDAR) to the representations, accordingly, when a new sensor type or modality is added, embodiments of the invention do not need to collect information from the new sensor type in order to use input from the new sensor type, rather, input from the new sensor type may be mapped to existing element representations already included in a database. This capability greatly improves the field since it enables introducing new sensor types automatically, without having to first collect information from a new sensor. As described, using machine learning, a mapping **330** between input from a first type of sensor to input from a second type of sensor is created such that given input from the first sensor type when sensing an element an embodiment can map input from a second type of sensor to a representation of the element. For example, system **200** may automatically create a new element representation **132** object based on an image (first sensor type) of pole **232** and, using mapping **330** (or a model as described) map input from a LIDAR (second sensor type, when it senses pole **323**) to the element representation **132** object that describes pole **323**. By mapping input from a plurality of different sensor types or modalities to an element representation **132** object embodiments of the invention enable using any sensor type or modality to identify elements, objects or features in a space.

(114) As further described, element representation **132** objects may be automatically created based on input from various or different sensors or modalities. An embodiment may continuously and automatically update element representation **132** objects, determine which element representations **132** are best used by different sensors or under different conditions. For example, if an embodiment determines that the color change feature **322** is not always identified then the embodiment may select to remove this feature from a databases or select not to use this feature, in another example, if an embodiment sees that house **321** is always sensed by a camera and a LIDAR then the embodiment may select to keep and use this object as described. Accordingly, embodiments of the invention may automatically and autonomously, without intervention or supervision of a human, create representations of elements in space and use the representations to determine a location as

described.
(115) In the description and claims of the present application, each of the verbs, “comprise” “include” and “have”, and conjugates thereof, are used to indicate that the object or objects of the verb are not necessarily a complete listing of components, elements or parts of the subject or subjects of the verb. Unless otherwise stated, adjectives such as “substantially” and “about” modifying a condition or relationship characteristic of a feature or features of an embodiment of the disclosure, are understood to mean that the condition or characteristic is defined to within tolerances that are acceptable for operation of an embodiment as described. In addition, the word “or” is considered to be the inclusive “or” rather than the exclusive or, and indicates at least one of, or any combination of items it conjoins.
(116) Descriptions of embodiments of the invention in the present application are provided by way of example and are not intended to limit the scope of the invention. The described embodiments comprise different features, not all of which are required in all embodiments. Some embodiments utilize only some of the features or possible combinations of the features. Variations of embodiments of the invention that are described, and embodiments comprising different combinations of features noted in the described embodiments, will occur to a person having ordinary skill in the art. The scope of the invention is limited only by the claims.
(117) While certain features of the invention have been illustrated and described herein, many modifications, substitutions, changes, and equivalents may occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the invention.
(118) Various embodiments have been presented. Each of these embodiments may of course include features from other embodiments presented, and embodiments not specifically described may include various features described herein.

Claims

1. A computer-implemented method of determining at least one of a location and an orientation of a sensor, the computer-implemented method comprising: including, in a primary database (**131**), representations of elements in a first space; using an artificial intelligence engine that produces a machine learning model adapted to extract or predict, from input obtained from (i) a first perception sensor (**310**) or (ii) a first modality, one or more of location indicative elements of said elements that are characteristic or defining of a geographical location or position; creating a mapping between (i) the representations of the elements including said one or more location indicative elements in the first space and (ii) the input from the first perception sensor or the first modality; utilizing, in a second space, (i) the mapping and (**11**) input from a second perception sensor or a second modality that is different from the first perception sensor or the first modality, to identify one or more of the elements, including one or more of the one or more location indicative elements, whose representations are included in the database; and using attributes of the identified one or more of the elements including said one or more location indicative elements to determine at least one of a location and an orientation of the second perception sensor or the second modality in the second space; wherein the computer-implemented method comprises: (a) automatically and autonomously creating said mapping of said space, based on input from a set of different sensors, by automatically creating in a primary database (**131**) the mapping of (**1**) said space and (ii) elements in said space, and automatically creating element representations (**132**) based on input from said set of different sensors, and automatically creating mapping between (I) element representations and (II) input from said set of sensors; wherein objects or features in said space are usable for determining a location and an orientation of a sensor; wherein after the mapping and element representations are created, input from a sensor can be mapped to element representations, and the element representations can be used to determine the location and the orientation of the

sensor; wherein step (a) comprises: storing in the primary database (**131**) a representation of an element in the first space; creating a mapping between (**1**) the representation of the element, and (ii) input from the first perception sensor (**310**): subsequently, using (I) said mapping and (II) input from the second perception sensor, in the second space, to locate or identify one or more elements in the primary database (**131**), (b) providing a positioning solution based on matching between features or objects that are captured by perception sensors to features or objects that are represented in said primary database (**131**), by matching between (**1**) an element as seen or captured from a first point-of-view, with (**11**) an element or feature as seen or captured from a second point-of-view; (c) correlating between (i) representations of elements that are included in said primary database (**131**) based on a top-view thereof, and (ii) elements that were captured or seen using street-view images, and matching elements in a top-view with elements in a ground surface view; (d) using auto-encoding features to learn (**1**) a first correlation between multiple Instances of a scene, and also (ii) a second correlation between multiple points of view, and also (iii) a third correlation between representations of a scene by a plurality of modalities; (e) using Machine Learning, creating a representation of an object based on input from a first type of sensor, and then identifying said object based on input from a second, different, type of sensor.

2. The computer-implemented method according to claim 1, comprising: based on an estimation of a location of a vehicle, downloading some of the element representations (**132**) from said primary database (**131**) to a vehicular database (**232**) of a specific vehicle (**230**), wherein the vehicular database (**232**) of the specific vehicle (**230**) only downloads from the primary database (**131**) and stores element representations (**132**) that describe elements that are physically located near the specific vehicle (**230**).

3. The computer-implemented method according to claim 1, comprising: at a localization Artificial Intelligence engine (**220**), training a Machine Learning model (**221**), by (i) receiving, at the localization Artificial Intelligence engine (**220**), from a dataset (**215**), data that includes images; wherein each data element (**218**) of said dataset (**215**) is associated with a geographical position and/or orientation; (ii) receiving, at the localization Artificial Intelligence engine (**220**), data originating from a camera and data originating from a LIDAR sensor; (iii) training, at the localization Artificial Intelligence engine (**220**), a Machine Learning model corresponding to data included in said dataset (**215**), and producing a Machine Learning model corresponding to a second type of data that originated from one or more sensors.

4. The computer-implemented method according to claim 3, comprising: (I) at the localization Artificial Intelligence engine (**220**), producing a Machine Learning model that is adapted to receive an input data element of a specific type of sensor; wherein the Machine Learning model is adapted to predict, from said received input data element, a location-indicative element (**320**) which is a characteristic of a geographical location; wherein the location-indicative element is unintelligible or incoherent to a human observer, and does not pertain to any specific physical object, and is a produce of the Machine Learning model (**221**) that can be recognized by the localization Artificial Intelligence engine (**220**) as indicative of geographical locations; (II) based on the Machine Learning model (**221**), producing, from an incoming data element, at least one road code (**250**) that is associated with said location-indicative element.

5. The computer-implemented method according to claim 1, comprising: adapting a radar to sense an object, and to output a data element corresponding to the sensed object; producing, by said radar, signals or data elements that correspond to a set of values or readings that represent a frequency or an amplitude or other attributes of reflected Radio Frequency radiation or electromagnetic waves; mapping said set of values to an element representation object; subsequently, when the set of values is received by another radar, mapping the set of values to said element representation object.

6. The computer-implemented method according to claim 1, wherein said mapping comprises generating a signature based on a set of values received from a sensor; wherein the method

comprises: utilizing said signature as a key for finding the correct element representation object in the vehicular database (**232**), by performing, at an Onboard Localization and Orientation Unit (**233**): (i) examining input from a sensor, (ii) producing a signature based on the input from the sensor, (iii) using said signature to find an element representation object in the vehicular database (**232**) of the specific vehicle (**230**); (iv) once the element representation object is found, extracting location attributes of the element from the element representation object.

7. The computer-implemented method according to claim 1, comprising: creating a first mapping according to a first condition, and creating a second mapping according to a second condition, and selecting to use one of the first mapping and the second mapping according to a condition.

8. The computer-implemented method according to claim 7, wherein the first condition is day time, wherein the second condition is night time.

9. The computer-implemented method according to claim 7, wherein the first condition is a rainy day, wherein the second condition is a sunny day.

10. The computer-implemented method according to claim 1, comprising: classifying element representations according to a condition, wherein elements that are best detected at night are associated with a first class, wherein elements that are best detected in sunlight are associated with a second class.

11. The computer-implemented method according to claim 1, comprising, comprising: automatically selecting a set of elements to be used for determining location and/or orientation, based on a condition; wherein, if it rains when location of a sensor (**311**) is to be determined, then selecting a set of elements that are classified as best for poor visibility for determining location and/or orientation of said sensor (**311**).

12. The computer-implemented method according to claim 1, comprising: classifying element representations according to a suitability for sensor type, wherein a first class of elements is suitable for use with a camera, wherein a second class of elements is suitable for use with a LIDAR sensor.

13. The computer-implemented method according to claim 1, comprising: creating a plurality of mappings (**330**) for a respective plurality of sensor types; and determining location and orientation of a sensor by selecting one of the plurality of mappings (**330**) based on the type of the sensor.

14. The computer-implemented method according to claim 1, comprising: (i) creating a plurality of mappings (**330**) for a respective plurality of sensor types; (ii) dynamically selecting a mapping to be used, based on the type of sensor for which location and/or orientation are determined; (iii) automatically selecting a subset of elements in the primary database (**131**), wherein the subset is selected based on the type of the sensor; (iv) selecting mappings and element representations by selecting the set of element representations (**132**) and mappings (**330**) that are downloaded to the vehicular database (**232**) of said specific vehicle (**230**) and are the optimal set for the sensor used; wherein the computer-implemented method comprises: selecting a subset of element representations (**132**) and mappings (**330**) based on: (I) sensor type, and (II) suitability of the sensor to sense specific elements, and (III) a condition; if input from the sensor cannot be mapped to any element representation (**132**), then: creating in the primary database (**131**) a new element representation (**132**).

15. The computer-implemented method according to claim 1, comprising: encoding (i) data received from a plurality of sensor types, to (ii) a unified format representation to be included in said primary database (**131**); and by encoding different data types coming from different sensor types into a unified format representation, representing a single element in space based on input from a plurality of different modalities.

16. A non-transitory storage medium having stored thereon instructions that, when executed by a machine, cause the machine to perform a method comprising: including, in a primary database (**131**), representations of elements in a first space; using an artificial intelligence engine that produces a machine learning model adapted to extract or predict, from input obtained from (i) a

first perception sensor (**310**) or (ii) a first modality, one or more of location indicative elements of said elements that are characteristic or defining of a geographical location or position; creating a mapping between (i) the representations of the elements including said one or more location indicative elements in the first space and (ii) the input from the first perception sensor or the first modality; utilizing, in a second space, (i) the mapping and (ii) input from a second perception sensor or a second modality that is different from the first perception sensor or the first modality, to identify one or more of the elements, including one or more of the one or more location indicative elements, whose representations are included in the database; and using attributes of the identified one or more of the elements including said one or more location indicative elements to determine at least one of a location and an orientation of the second perception sensor or the second modality in the second space; wherein the method comprises: (a) automatically and autonomously creating said mapping of said space, based on input from a set of different sensors, by automatically creating in a primary database (**131**) the mapping of (i) said space and (ii) elements in said space, and automatically creating element representations (**132**) based on input from said set of different sensors, and automatically creating mapping between (I) element representations and (II) input from said set of sensors; wherein objects or features in said space are usable for determining a location and an orientation of a sensor; wherein after the mapping and element representations are created, input from a sensor can be mapped to element representations, and the element representations can be used to determine the location and the orientation of the sensor; wherein step (a) comprises: storing in the primary database (**131**) a representation of an element in the first space; creating a mapping between (i) the representation of the element, and (ii) input from the first perception sensor (**310**); subsequently, using (I) said mapping and (II) input from the second perception sensor, in a the second space, to locate or identify one or more elements in the primary database (**131**) (b) providing a positioning solution based on matching between features or objects that are captured by perception sensors to features or objects that are represented in said primary database (**131**), by matching between (i) an element as seen or captured from a first point-of-view, with (ii) an element or feature as seen or captured from a second point-of-view; (c) correlating between (i) representations of elements that are included in said primary database (**131**) based on a top-view thereof, and (ii) elements that were captured or seen using street-view images, and matching elements in a top-view with elements in a ground surface view; (d) using auto-encoding features to learn (i) a first correlation between multiple instances of a scene, and also (ii) a second correlation between multiple points of view, and also (iii) a third correlation between representations of a scene by a plurality of modalities; (e) using Machine Learning, creating a representation of an object based on input from a first type of sensor, and then identifying said object based on input from a second, different, type of sensor.

17. A system comprising: one or more hardware processors that are configured to execute code, and that are operably associated with a memory unit that is configured to store code and data, wherein the one or more hardware processors are configured to perform a method comprising: including, in a primary database (**131**), representations of elements in a first space; using an artificial intelligence engine that produces a machine learning model adapted to extract or predict, from input obtained from (i) a first perception sensor (**310**) or (ii) a first modality, one or more of location indicative elements of said elements that are characteristic or defining of a geographical location or position; creating a mapping between (i) the representations of the elements including said one or more location indicative elements in the first space and (ii) the input from the first perception sensor or the first modality; utilizing, in a second space, (i) the mapping and (ii) input from a second perception sensor or a second modality that is different from the first perception sensor or the first modality, to identify one or more of the elements, including one or more of the one or more location indicative elements, whose representations are included in the database; and using attributes of the identified one or more of the elements including said one or more location indicative elements to determine at least one of a location and an orientation of the second

perception sensor or the second modality in the second space; wherein the method comprises: (a) automatically and autonomously creating said mapping of said space, based on input from a set of different sensors, by automatically creating in a primary database (**131**) the mapping of (i) said space and (ii) elements in said space, and automatically creating element representations (**132**) based on input from said set of different sensors, and automatically creating mapping between (I) element representations and (II) input from said set of sensors; wherein objects or features in said space are usable for determining a location and an orientation of a sensor; wherein after the mapping and element representations are created, input from a sensor can be mapped to element representations, and the element representations can be used to determine the location and the orientation of the sensor; wherein step (a) comprises: storing in the primary database (**131**) a representation of an element in the first space; creating a mapping between (i) the representation of the element, and (ii) input from the first perception sensor (**310**); subsequently, using (I) said mapping and (II) input from the second perception sensor, in the second space, to locate or identify one or more elements in the primary database (**131**), (b) providing a positioning solution based on matching between features or objects that are captured by perception sensors to features or objects that are represented in said primary database (**131**), by matching between (i) an element as seen or captured from a first point-of-view, with (ii) an element or feature as seen or captured from a second point-of-view; (c) correlating between (i) representations of elements that are included in said primary database (**131**) based on a top-view thereof, and (ii) elements that were captured or seen using street-view images, and matching elements in a top-view with elements in a ground surface view; (d) using auto-encoding features to learn (i) a first correlation between multiple instances of a scene, and also (ii) a second correlation between multiple points of view, and also (iii) a third correlation between representations of a scene by a plurality of modalities; (e) using Machine Learning, creating a representation of an object based on input from a first type of sensor, and then identifying said object based on input from a second, different, type of sensor.
